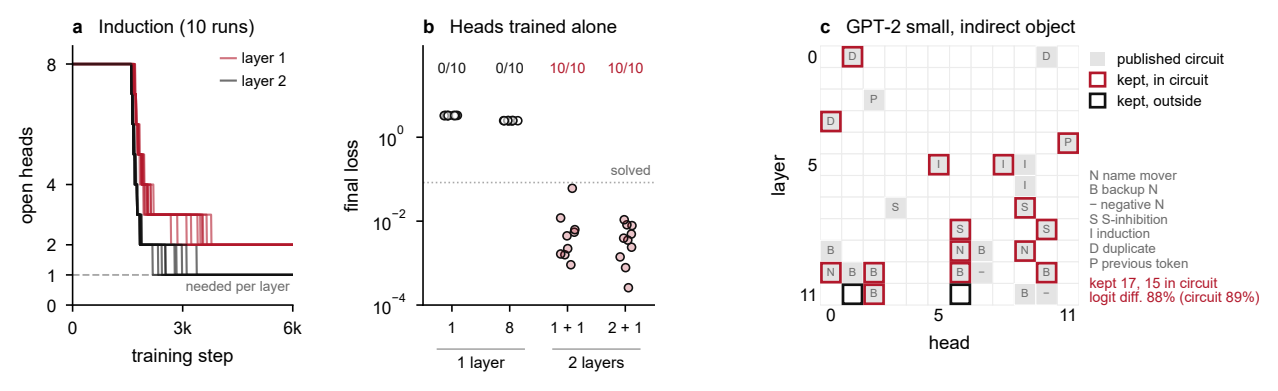
a Induction (10 runs)
layer 1
layer 2
open heads
8
4
2
1
0
needed per layer
0
3k
6k
training step
b Heads trained alone
0/10
0/10
10/10
10/10
final loss
10^0
10^{-2}
10^{-4}
solved
1
8
1 + 1
2 + 1
1 layer
2 layers
c GPT-2 small, indirect object
published circuit
kept, in circuit
kept, outside
layer
head
0
5
11
N name mover
B backup N
− negative N
S S-inhibition
I induction
D duplicate
P previous token
kept 17, 15 in circuit
logit diff. 88% (circuit 89%)